

L1.6 Trig & the Chain Rule

Recall: L1.5 Differentiation Formulae

Name _____

Solo – 9 minutes

1. Differentiate.

(a) $y = (x^2 + 3)(2x - 1)$

(b) $y = \frac{2x + 1}{x^2 + 1}$

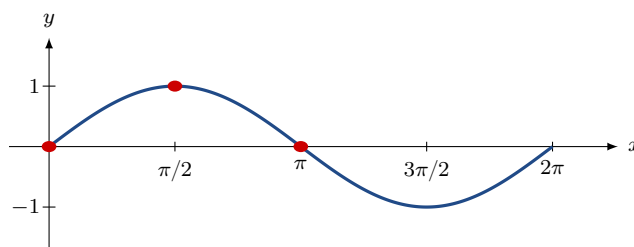
2. What rules are needed to compute the derivative in the easiest way possible?

$$\frac{x^4 + x}{x^2}$$

$$x^3(x^2 - 5)$$

$$\frac{x + 1}{x - 1}$$

3. The graph of $f(x) = \sin x$ is below, with three points marked.



(a) Read the **slope of the curve** at each marked point and fill in the table.

x	0	$\pi/2$	π
slope			

(b) What familiar function has those three values at those three inputs?